Home Lab Project Week 10, Day 2

Test the firewall rules using controlled connectivity tests from Kali Linux and the target environment. Document allowed and blocked traffic and correct configuration issues.

This is the verification part of the assignment. You'll prove each allowed path works, prove each blocked path is dropped **by the firewall**, record the results, and fix anything that doesn't match.

The examples below assume the planned layout: **Kali `192.168.10.10`** on LAN and **Ubuntu in the DMZ**, e.g. `192.168.20.10` with gateway `192.168.20.1`. Replace these with your real IPs. If you haven't built the DMZ yet, you can run the same method against your practice rules.

Step 1: Prepare before testing

A good test isolates one variable: the firewall. First rule out everything else.

On OPNsense:

1. Confirm your rules are saved, in the right order, and you clicked **Apply**.
2. Confirm the "Default allow LAN to any" rules are **disabled**.
3. Go to **Firewall → Diagnostics → States** and flush all states.
4. Open **Firewall → Log Files → Live View** in a separate tab and keep it open.

On Ubuntu (the target):

1. Make sure Ubuntu's own firewall isn't interfering. If ufw is active, any block could come from Ubuntu instead of OPNsense.

bash
```
sudo ufw status
```

You want `Status: inactive`.
2. Confirm the services are running and listening:

bash
```
sudo systemctl status ssh apache2 --no-pager
sudo ss -tlnp | grep -E ':22 |:80 |:8080 '
```

3. Start the **decoy** service in a second terminal and leave it running:

bash
```
python3 -m http.server 8080
```

4. Check that the gateway is OPNsense:

bash
```
ip route
```

`default via 192.168.20.1` is correct.

The decoy on port 8080 is the key to the whole test. It's a real service that **is** listening. If Kali can't reach it, the only explanation is that the firewall blocked it.

Step 2: Tests from Kali (the source)

Run these one at a time and screenshot each result.

Should be allowed

ID	Test	Command	Expected result
A1	Ping	`ping -c 4 192.168.20.20`	4 replies, 0% loss
A2	SSH port	`nc -zv -w 3 192.168.20.20 22`	`succeeded` / `open`
A3	SSH login	`ssh <user>@192.168.20.20`	Password prompt
A4	HTTP	`curl -I -m 5 http://192.168.20.20`	`HTTP/1.1 200 OK`

Should be blocked

ID	Test	Command	Expected result
B1	Decoy service	`nc -zv -w 3 192.168.20.20 8080`	**Timed out**
B2	HTTPS	`nc -zv -w 3 192.168.20.20 443`	**Timed out**
B3	FTP	`nc -zv -w 3 192.168.20.20 21`	**Timed out**
B4	MySQL	`nc -zv -w 3 192.168.20.20 3306`	**Timed out**

Summary scan

bash
```
nmap -Pn -p 21,22,80,443,3306,8080 192.168.20.20
```

Expected: **22 open, 80 open**, and everything else **filtered**. This one screenshot shows your whole policy.

Step 3: Tests from Ubuntu (the target environment)

These check that the server can't start connections back to the attacker box.

ID	Test	Command	Expected result
B5	Ubuntu → Kali ping	`ping -c 4 192.168.10.10`	**No replies**
B6	Ubuntu → Kali SSH	`nc -zv -w 3 192.168.10.10 22`	**Timed out**

Prove the packets never arrived

While Kali repeats test **B1**, run this on Ubuntu:

bash
```
sudo tcpdump -n -i any port 8080
```

If **no packets appear**, OPNsense dropped them before they reached Ubuntu. Repeat with `port 80` during test A4, and you should **see** packets. This side-by-side comparison is strong evidence. If tcpdump is missing, install it with `sudo apt install tcpdump`.

Step 4: Firewall-side evidence

1. In **Live View**, filter by Kali's IP. Blocked tests (B1–B4) should appear as red **block** entries naming your deny rule.
2. In **Firewall → Diagnostics → States**, allowed tests (A1–A4) appear as active sessions between Kali and Ubuntu.

How to read results

Result you see	Meaning
`succeeded` / `200 OK` / ping replies	Traffic allowed and the service answered

Timed out	Firewall silently dropped it (a correct block)
Connection refused	Traffic got **through** the firewall, but nothing is listening on that port. It's not a firewall block.
`No route to host`	Routing or gateway problem, not a rule problem

For a blocked test, "connection refused" is a warning sign: the firewall let it through.

Step 5: Record your results

Fill in this table for your documentation:

ID	Source → Dest	Protocol/Port	Expected	Actual	Evidence	Pass/Fail
A1	Kali → Ubuntu	ICMP	Allow		Screenshot #	
A2	Kali → Ubuntu	TCP 22	Allow			
A4	Kali → Ubuntu	TCP 80	Allow			
B1	Kali → Ubuntu	TCP 8080	Block			
B2	Kali → Ubuntu	TCP 443	Block			
B5	Ubuntu → Kali	ICMP	Block			

A **Fail** is fine as long as you fix it and document how. That's the "correct configuration issues" part of the assignment.

Step 6: Fix configuration issues

Allowed traffic is blocked

Likely cause	How to check	Fix
Rule on the wrong interface	Rules must be on **LAN**, where Kali's traffic **enters**	Move or recreate the rule on LAN
Block rule above pass rule	Look at rule order	Move the pass rule above the block
Alias has the wrong IP	**Firewall → Diagnostics → Aliases**	Edit the alias, then **Apply**

Ubuntu gateway isn't OPNsense	`ip route` on Ubuntu	Set the gateway to `192.168.20.1`
Ping fails, TCP works	Is there an ICMP rule?	Add a Pass rule for ICMP (Echo Request)
"Connection refused"	`systemctl status ssh apache2`	Start the service (not a firewall issue)

Blocked traffic is allowed

Likely cause	How to check	Fix
Default allow rule still on	Rule shows ☑ / green ▸	Disable it, then **Apply**
Old connection still open	**Diagnostics** → **States**	Flush states and retest
Hosts on the same subnet	`ip a` on both	Finish the DMZ setup
Port alias too broad	Check the alias contents	Keep only 22, 80

Document each fix

Use this format for every issue you correct:

> **Issue:** Kali could not SSH to Ubuntu (A2 failed, timed out).
> **Root cause:** The pass rule was placed below the deny-all rule.
> **Fix:** Moved the pass rule above the deny rule, clicked Apply, flushed states.
> **Retest:** A2 succeeded.

An honest issue → cause → fix → retest record often scores higher than a perfect first run, because it shows troubleshooting skill.

Once you've run the tests, would you like me to turn your results and screenshots into a formatted lab report doc for Week 9 Day 4?